

Cambridge Assessment
International Education

Question	Answer	Marks
1(a)	H_2S (1)	**1**
1(b)	MnO_4^- (1)	**1**
1(c)	NH_4^+ (1)	**1**

Question	Answer	Marks
2(a)(i)	0.155 – 0.190 (nm) (1)	**1**
2(a)(ii)	the melting point goes up and down (1)	**1**
2(b)	has **many** strong bonds (that have to be broken or overcome) / needs lots of energy to break or overcome **many** bonds (1)	**1**
2(c)(i)	aluminium oxide (dissolved) in (molten) cryolite (1)	**1**
2(c)(ii)	carbon / graphite (1)	**1**
2(c)(iii)	negative electrode: $Al^{3+} + 3e^- \rightarrow Al$ (1) positive electrode: $2O^{2-} \rightarrow O_2 + 4e^-$ (1)	**2**
2(d)	Al because it loses electrons (1)	**1**
2(e)	has a layer of oxide / aluminium oxide layer (1) layer is impermeable (to water) / coating is impermeable (to water) (1)	**2**

Question	Answer	Marks
2(f)	use hydrochloric acid (1) use excess **aluminium** (1) filter (off aluminium) (1) leave filtrate in warm place / evaporate solution to point of crystallisation then leave / leave in the sun (1)	**4**

Question	Answer	Marks
3(a)	high melting point / high boiling point / high density / (good) conductor of electricity / (good) conductor of heat / malleable / ductile / hard / strong / sonorous (1)	**1**
3(b)(i)	simple covalent molecule / simple covalent molecular (1)	**1**
3(b)(ii)	$TiCl_4(l) + 2H_2O(l) \rightarrow TiO_2(s) + 4HCl(aq)$ balanced equation (1) state symbols – dependent on correct formulae (1)	**2**

Question	Answer	Marks
3(c)	**Process 1:** mol $TiCl_4$ = $\frac{1000}{190}$ **OR** 5.263 (1) **Process 2:** mass Ti = $\frac{1000}{190} \times 48$ **OR** 252.63 (1) **Process 3:** % = $\frac{1000}{190} \times 48 \times 0.9$ **OR** 227.36 / 230g **OR** **Process 1:** mol $TiCl_4$ = $\frac{1000}{90}$ **OR** 5.263 (1) **Process 2:** % = $\frac{90}{100} \times 5.263$ **OR** 4.736 (1) **Process 3:** mass = 4.736×48 **OR** 227.36 / 230 g	**3**

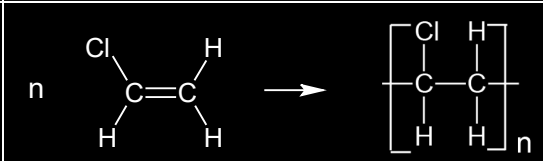
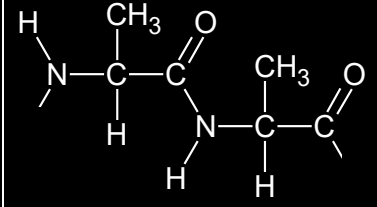
Question	Answer	Marks
4(a)	One mark each for any three of: • fractional distillation • **petroleum** is heated / **petroleum** vaporised (into column) • idea of components / fractions with different boiling points • (fractions with) low boiling point come off at the top / (fractions with) high boiling point come off at the bottom / fractions come off at different levels (in the column)	**3**
4(b)	(surfacing) roads / (water-proofing) roofs (1)	**1**
4(c)(i)	idea that the formula fits C_nH_{2n+2} e.g. for $C_{12}H_{26}$ n=12 and 2n+2 is 26 (1)	**1**
4(c)(ii)	correct balanced equation e.g. $C_{12}H_{26} \rightarrow C_{12}H_{24} + H_2$ **OR** $C_{12}H_{26} \rightarrow H_2 + 3C_4H_8$ etc. (1)	**1**
4(d)	(hydrogen) does not form carbon dioxide (1) so does not lead to greenhouse effect / flooding / drought / climate change (1) **OR** water is the **only** product (1) so does not lead to greenhouse effect / flooding / drought / climate change (1) **OR** (hydrogen) renewable / sustainable (1) hydrogen is not finite (resource) / (the product water) can be split up (or electrolysed) to give hydrogen / hydrogen can be formed again from water (1)	**2**

<table>
<tr><th>Question</th><th>Answer</th><th>Marks</th></tr>
<tr><td>5(a)</td><td>
<table>
<tr><td>element</td><td>C</td><td>H</td><td>O</td></tr>
<tr><td>%</td><td>57.1</td><td>4.8</td><td>38.1</td></tr>
<tr><td>moles</td><td>4.76</td><td>4.8</td><td>2.38</td></tr>
<tr><td>simplest mole ratio</td><td>2</td><td>2</td><td>1</td></tr>
</table>
% of oxygen (1)
moles / mole ratio (1)
empirical formula C_2H_2O (1)
</td><td>3</td></tr>
<tr><td>5(b)</td><td>moles of KOH = 0.0185 × 0.250 OR 0.004625 (1)
moles of W = moles of KOH ÷ 3 OR 0.004625 ÷ 3 OR 0.00154 (1)
M_r = (0.194 / 0.00154) = 125.97 / 126 (1)</td><td>3</td></tr>
<tr><td>5(c)</td><td>$C_6H_6O_3$ (1)</td><td>1</td></tr>
</table>

Question	Answer	Marks
6(a)	removal of salt(s) (from sea water) (1)	**1**
6(b)(i)	fertilisers (1)	**1**
6(b)(ii)	eutrophication (1)	**1**
6(c)(i)	filtration (1)	**1**
6(c)(ii)	carbon / charcoal (1)	**1**
6(c)(iii)	chlorine (1)	**1**

Question	Answer	Marks
7(a)	ammonium iodide disappears / solid disappears / sample disappears / nothing left in tube (1)	**1**
7(b)	moles of ammonium iodide $\frac{2.9}{145}$ **OR** 0.02(00) (1) moles of gas = 2×0.02 **OR** 0.04(0) (1) volume of gas = $(0.04 \times 24) = 0.96\ dm^3$ **OR** $960\ cm^3$ (1)	**3**
7(c)	(acidified aqueous) silver nitrate (1) (pale) yellow ppt (1)	**2**
7(d)	$2I^-(aq) + Br_2(aq) \rightarrow I_2(aq) + 2Br^-(aq)$ correct formulae and balanced (1) correct state symbols – dependent on correct formulae (1)	**2**
7(e)	in solid ions cannot move (1) in aqueous / solution ions can move (1)	**2**

Question	Answer	Marks
8(a)	idea that no reactants or products can escape (1)	**1**
8(b)	Less NO_2 / concentration of NO_2 decreases / more NO / more O_2 (1) fewer moles **of gas** on the right hand side of the equation / fewer **gas** molecules on the right hand side of the equation (1)	**2**
8(c)(i)	the (forward) reaction releases heat / the (forward) reaction is exothermic (1)	**1**
8(c)(ii)	particles have less **kinetic** energy / particles moving slower (1) less successful collisions / less energetic collisions / less effective collisions / less particles with equal or above activation energy (1)	**2**
8(d)	correct structure (1)	**1**
8(e)	$2NO_2 + H_2O \rightarrow HNO_3 + HNO_2$ (1)	**1**
8(f)	strong acid completely dissociates / strong acid completely ionises (1) weak acid partially dissociates / weak acid partially ionises / little dissociation (1)	**2**

Question	Answer	Marks
9(a)	 correct monomer (1) correct use of n **in front** of monomer to balance the equation (1)	**2**
9(b)(i)	**2**C_2H_3Cl + **5**O_2 → **4**CO_2 + **2**H_2O + **2**HCl (1)	**1**
9(b)(ii)	HCl is an acid(ic gas) (1)	**1**
9(c)(i)	clothing / sleeping bags / rope / parachutes / fishing line / curtain materials (1)	**1**
9(c)(ii)	 (1)	**1**
9(d)	**liquid** / it changes from moving to vibrating / particles sliding over each other (in liquid) to vibrating (in solid) (1) **liquid** / it changes from disordered to ordered (in solid) / (particles) changing from irregularly arranged (in liquid) to regularly arranged (in solid) (1)	**2**
9(e)	positive ions in a sea of electrons (1) strong attraction between electrons and (positive) ions (1)	**2**

Question	Answer	Marks
10(a)	contains oxygen (atom) (1)	**1**
10(b)	does not have any carbon-carbon double bonds / **all** the carbon-carbon bonds are single bonds / the carbon-carbon bonds are **only** single bonds (1)	**1**
10(c)	**EITHER** test: Universal indicator (1) cyclobutanol: goes green / does not change colour (1) butanoic acid: goes red / orange or yellow (1) **OR** test: heat with named alcohol (1) cyclobutanol: no reaction (1) butanoic acid: gives sweet smelling compound (1) **OR** test: heat with named carboxylic acid (1) cyclobutanol: gives sweet smelling compound (1) butanoic acid: no reaction (1)	**3**
10(d)	M_r = 72 (1) % = 66.7 / 67 (1)	**2**

Question	Answer	Marks
10(e)(i)	ethanoic acid / CH_3COOH (1)	**1**
10(e)(ii)	(acidified) potassium manganate(VII) (1)	**1**
10(e)(iii)	methanol (1)	**1**